

AI Grand Prix Physical Qualifier Technical Specification

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1. Document Control

1.1 Revision History

Issue	Date	Author	Summary of Changes
00.01	2026-08-18	NT	First release
00.02	2026-08-03	NT	First release

1.2 Audience

This specification is addressed to competition participants (“Teams”). It defines the technical interface and operational requirements required to develop autonomous control software for the physical qualifier AI Drone Race.

2. Purpose and Scope

2.1 Purpose

This document provides contestants with essential information regarding the physical qualifier event, the racing environment, and the provided drone and compute hardware. Additionally, it details the software stack specifications.

2.2 Scope

- Rules
- On site program
- Specs of the software stack
- Track environment definition
- qualification run requirements

2.3 Out of Scope

- internal simulator architecture
- event operations
- contractual or commercial matters

3. Physical Qualifier

3.1 Objective

Validate autonomous drone flight capabilities through a timed obstacle course, qualifying teams based on navigation precision, gate passage count, and lap performance.

3.2 Program

On site teams will have dedicated slots every day to train the software performance in the real track

Time Slots: Multiple defined operational slots per team per day.

Timing Activation: Runs are classified and timed starting from the first timing line cross / gate trigger.

3.3 Qualification rules

Ranking Criteria: Leaderboard ranking is strictly determined based on the timing results of valid, completed runs.

In order a run being valid the drone must fly without any intervention of human pilots

Run Requirements:

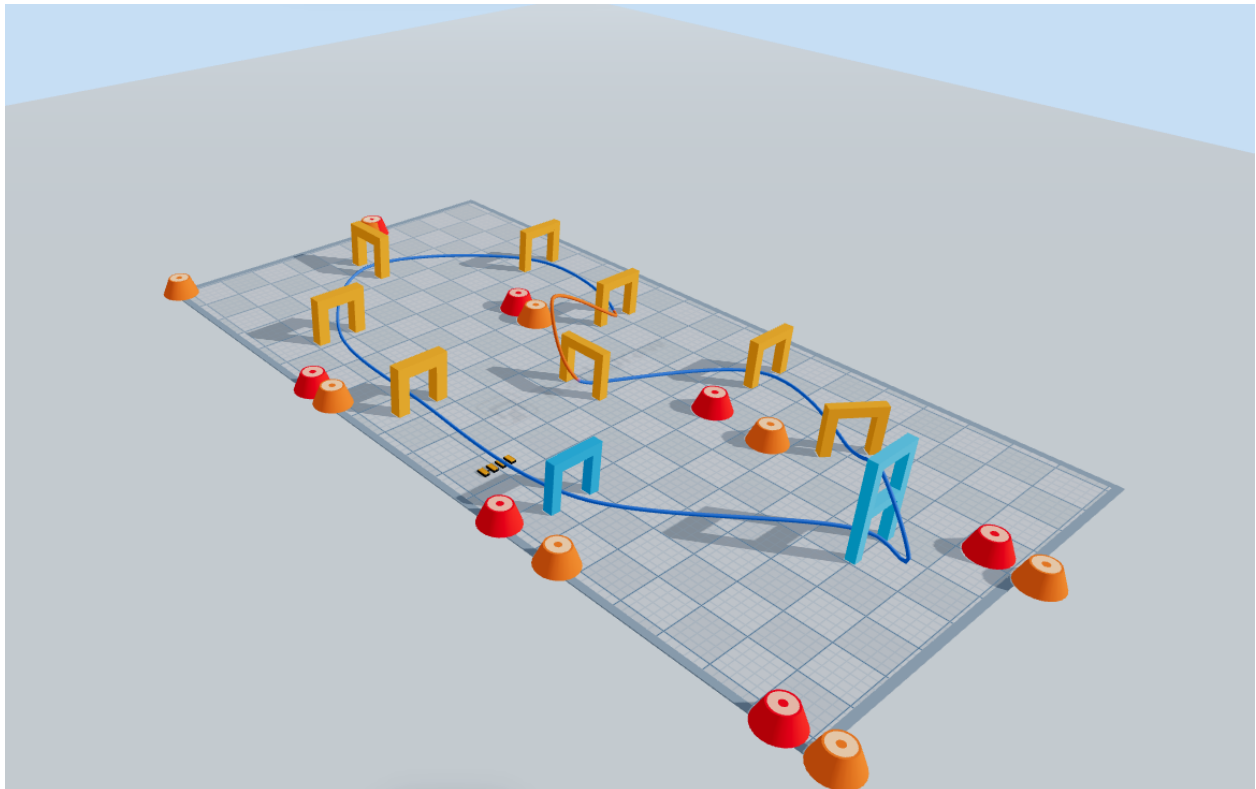
- In order to complete a run, the teams must complete 2 laps of the track.

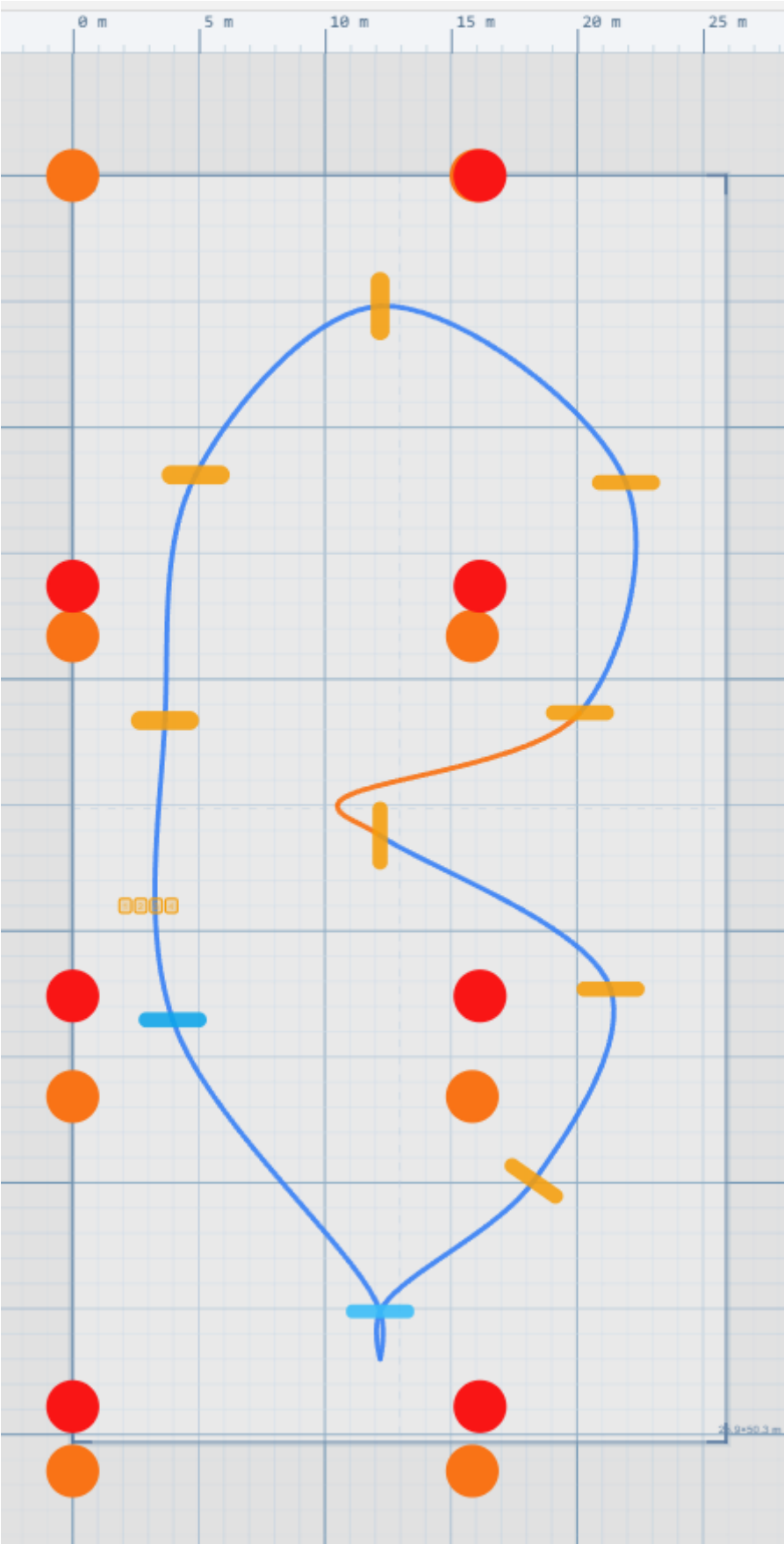
Qualifications requirement:

- **Qualification & Scoring**
Evaluated by the best overall time recorded during the final two days
- **Primary Scoring:**
Ranked by the maximum number of successfully passed gates.
- **Tie-Breaker:**
Fastest elapsed time to the last validly crossed gate.

3.4 Track

- **Footprint:** 60m x 21m
- The track consists of multiple gate gates including a double gate

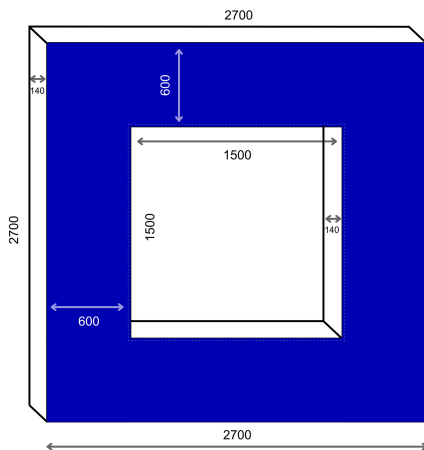




3.5 Gates

Gate boundaries

- Width: 2700mm
- Height: 2700mm
- Depth: 80mm



Gate inner square boundaries:

- Width: 1500mm
- Height: 1500mm
- Depth: 80mm

3.6 Drone

- **Platform:** Archer B2 Frame
- **Flight Controller (FC):** Running Betaflight firmware.
- **Inter-Board Protocol:** Serial UART protocol for high-speed telemetry and command communication with the AI compute module.
- The drone uses 8-inch propellers with a 4.1 pitch. Full details on motors, ESCs, battery specs, IMU hardware, and total all-up weight (AUW). More details can be provided on site.

3.7 Compute Unit

Hardware:

- **System-on-Module:** NVIDIA Jetson Orin NX (16 GB)
- **Carrier / Breakout Board:** Seeed Studio A603 Carrier Board

Software Stack:

- **Base SDK:** NVIDIA JetPack 6.2
- **Provided Software:** Demo applications for camera sensor bring-up and FC-to-AI board serial communication.

3.8 Sensors

- The camera uses a rolling shutter sensor running at 1920×1080 resolution at 60 FPS (Mode , with a Horizontal Field of View (HFOV) of 75° .
- **Vision System:** Arducam 12.3 MP HQ Camera Module with M12 Lens Mount
- IMU is calibrated by Betaflight upon power on and provides IMU data at 50 Hz.

4. FAQ

- **What is our access level to Betaflight?**

You can configure rates, PIDs, filters, and telemetry via the Betaflight CLI. However, full reflashing of the flight controller is not allowed.

- **How do the Flight Controller (FC) and Nvidia Jetson communicate?**

They connect via UART using MSP. The interface transmits RC control signals (TX) and streams

- **What are the Jetson system specs, power modes, and privileges?**

You have full root access on the Jetson Orin NX. The Jetson operates in a 25W power mode, and a dedicated power supply will be provided in your workshop area.

- **How are camera frames timestamped?**

Frames are timestamped directly on the Jetson Orin NX using `CLOCK_MONOTONIC`.

- **Can we control camera settings and get intrinsic/extrinsic calibrations?**

You can control exposure and gain. Formal camera calibration matrices (intrinsic/extrinsic) will not be provided, but the physical position and orientation of the IMU relative to the drone will be supplied.

- **Will you provide goggles and a controller for flying the drone?**

A ground station and RC controller will be provided for each team when needed.

- **Can we adjust camera tilt during flight?**

Yes, pitch adjustments can be made during flight.

- **Are the control APIs in VQ2 based on the actual drone, or is there an additional transition layer?**

The APIs will differ from the physical drone setup. Full details on transition calls and timing will be provided in future documentation.

- **Can we issue an emergency shutdown?**

You have a manual override function via the RC controller that immediately supersedes autonomous functions at any time.

- **Can camera footage be transmitted mid-flight?**

Yes, the drone includes an onboard video transmitter (VTX).

- **Will teams receive training upon arrival?**

Yes, setup instructions and hands-on support for drone arming will be provided. Spectator access and future qualifier schedules will be communicated at a later date.

- **What actions result in disqualification?**

Any non-compliance or action outside the official rulebook results in immediate disqualification.

- **Can we stand near the flight path with PPE?**

No, standing near the flight path is strictly prohibited regardless of PPE.

- **What happens if a drone gets damaged during flight?**

Teams are not financially liable for drone damage. All repairs, maintenance, and battery charging are managed on-site by the Neros repair team. Teams may not use personal battery chargers.

- **Will a course map or external tracking systems (like motion capture) be available?**

No external motion trackers or true-state tracking systems exist inside the building. Course map distribution details will be announced later.

- **How much flight time does each team get?**

Teams receive dedicated slots per day per team, alongside dedicated time in the training cage. Manual piloting is permitted.

- **Do teams share drones or data?**

No, data is not shared among teams. Each team is assigned a fleet of 4 dedicated drones.